CHEM 515 Practice Session 3: Vibrational Frequency Analysis and Chemical Reaction Modeling

Name: ________________________

In this practice, you will use the DFT methods available in Q-Chem to (i) compute harmonic frequencies and IR spectrum of the CS_2 molecule; (ii) locate the transition state and calculate the reaction (free) energy barrier for a Diels-Alder reaction. The calculations involved in **Exercise 1** can all be completed using the IQmol server, while for **Exercise 2**, you should use the Itcotck server to perform all the calculations.

You will need to **complete the tables and short-answer questions on this handout** and also prepare the requested files for submission. Please refer to the **Assignment page** on Canvas for detailed submission guidelines. **8 points** are associated with the completion of this handout and the rest **2 points** are for attendance.

Exercise 1: Vibrational frequencies of the CS_2 molecule

In this practice, you will compute the harmonic vibrational frequencies of the CS_2 molecule at two different levels of theories: (i) B3LYP/6-31+G(d) and (ii) r2SCAN/def2-TZVP. You will then compare the results to theoretical references and experimental fundamental frequencies. You will then generate a computed IR spectrum using one of these methods.

While the *theoretical best geometry* ("TBG") of CS_2 is provided, vibrational frequency analyses are most commonly performed using the geometries optimized at the same level of theory as for the frequency analysis ("OptG"). Let's start with obtaining the "OptG" frequencies using B3LYP/6-31+G(d):

- Open "`CS2_TBG.xyz`". Set up a geometry optimization calculation ("`Calculate` → `Geometry`") using the B3LYP functional and the 6-31+G(d) basis (equivalent to `6-31+G*`).
- Click the green "`+`" button to add a second job. The `$molecule` section should just contain "`read`". Now set "`Calculate` → `Frequencies`". Then under "SCF Control", change "`Guess`" to "`READ`", which will read in the converged SCF solution for the last (optimized) geometry from the first job.
- Submit the job. Once finished, you can find the computed harmonic frequency for each mode under "`Frequencies`" on the left column of the IQmol window ("Model View"). Report the frequencies under "OptG" for the corresponding method in **Table 1**. **(0.25 pt)**

Repeat the same procedure with "`METHOD = r2SCAN`" and "`BASIS = def2-TZVP`" and report the obtained frequencies in **Table 1**. **(0.25 pt)** Now, visualize the vibrational mode corresponding to each frequency based on the r2SCAN/def2-TZVP results (you just need to double-click each frequency under Model View). **Describe the vibrational motions** that are of the *lowest* and *highest* frequencies: **(0.3 pt)**

The definition of *root-mean-square error* (RMSE) is as follows:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \tag{1}$$

where each i corresponds to an individual data point and N is the total number of data points; y_i and $\hat{y}_i$ are the observed and true values for each data point, respectively. Using this formula, calculate the RMSEs

of the "OptG" frequencies given by those two methods, with each measured against the *theoretical reference values for harmonic frequencies* (the last column of **Table 1**). Report the two "OptG" RMSEs in the last row of **Table 1**. **(0.2 pt)**
Note: (i) consider ω_1 and ω_2 as one data point as they are degenerate modes; (ii) you should try to do the RMSE calculation using Excel.

Table 1: Harmonic frequencies of CS_2 (in cm^{-1}) calculated using B3LYP/6-31+G(d) and r2SCAN/def2-TZVP at the optimized geometries ("OptG") and the theoretical best geometries ("TBG"). The DFT frequencies are *unscaled* and compared to the high-level reference values for the harmonic frequencies

	B3LYP/6-31+G(d)		r2SCAN/def2-TZVP		Theor. Ref.
	OptG	TBG	OptG	TBG	
$\omega_{1,2}$					400.0
ω_3					674.1
ω_4					1562.7
RMSE					—

In the DFT benchmark paper by Liang and Head-Gordon et al. discussed in class, it was suggested that the errors in DFT's prediction of harmonic frequencies may be reduced by using the theoretical best geometries (TBGs). Let's test this idea here. Since the TBG for CS_2 has been provided ("`CS2_TBG.xyz`"), to get the results you only need to set up frequency calculations using the two levels of theory *without* optimizing the geometry first.

Report the TBG frequencies calculated using B3LYP/6-31+G(d) and r2SCAN/def2-TZVP in the corresponding columns in **Table 1** and calculate the RMSEs. **(0.25 pt)** Then, based on the RMSEs, **discuss** whether using the TBG of CS_2 improves the accuracy of the harmonic frequency predictions. **(0.25 pt)**

As introduced in the lecture, the calculated harmonic frequencies need to be *scaled* to make them more comparable to experimental frequencies. The scaling factor for B3LYP/6-31+G(d) can be found on the CCCBDB website (note: 6-31+G(d) is identical to 6-31+G(d,p) for C and S atoms), while the scaling factor for r2SCAN/def2-TZVP has been provided in **Table 2**.

Complete Table 2 by filling in the scaling factor for B3LYP/6-31+G(d) as well as the *scaled* frequencies of the two methods evaluated using OptG. Then **calculate** the RMSEs relative to the experimental reference values. **(0.4 pt)** Based on the RMSEs, answer the following questions:

1. With the scaling factor applied, which method works better in predicting the experimental frequencies? You should refer to data to support your answer. **(0.2 pt)**

2. What would be the optimal scaling factor for B3LYP/6-31+G(d) if we only consider the vibrational modes of CS_2? You can obtain this value by performing a least-squares fit of the calculated OptG frequencies (x) against the experimental references (y) as $y = ax$ (the intercept should be set to 0). Then, with the updated scaling factor, what is the RMSE of the scaled B3LYP/6-31+G(d) frequencies? **(0.4 pt)**

Table 2: Scaled DFT frequencies (in cm^{-1}, evaluated at OptG) compared to the experimental values

	B3LYP/6-31+G(d)	r2SCAN/def2-TZVP	Expt. Ref.
Scaling factor		0.967	—
$\omega_{1,2}$			397.0
ω_3			658.0
ω_4			1535.0
RMSE			—

As the final task in **Exercise 1**, you are supposed to generate an IR spectrum using the *scaled* r2SCAN/def2-TZVP (OptG) frequencies. Steps to follow:

- Under `Model View` for the corresponding output, double-click on "`Frequencies`", a new window with title "Vibrational Frequencies" should pop up.
- On the left bottom of the window, set the `Scale Factor` to be the same as in **Table 2**.
- Under the generated spectra on the right of the window, select the "`Lorentzian`" line shape. Now you should see the spectral lines are broadened.
- Save the spectrum as an image. **Submit it with your report**. **(0.2 pt)**

Looking at calculated IR spectrum for CS_2, how many peaks are present? Which vibrational mode shows no IR signal? Explain why that is the case. **(0.3 pt)**

Exercise 2: Predicting the reaction energy and activation barrier for a Diels-Alder reaction

In this exercise, you will be calculating the reaction free energy and the free energy barrier for the Diels-Alder (D-A) reaction between **ethene** and **cyclopentadiene**, which produces **norbornene**:

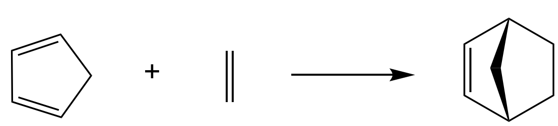

Figure 1: The D-A reaction to be investigated

Before getting to the last part of this exercise (for **Table 4**), *all* the calculations involved should be performed at the **B3LYP-D3(BJ)/def2-SVP** level. You should use the Itcotck server to perform these calculations. Please *do not* forget to add the "–D3(BJ)" correction (on IQmol it's added by "`Advanced` → `DFT` → `Dispersion Correction` → `DFT-D3(BJ)`"). Note that here a small basis set (`def2-SVP`) is employed to allow the jobs to finish quickly. In real production calculations, a larger basis set should be used for this relatively small system.

Table 3: B3LYP-D3(BJ)/def2-SVP results for the reactant, product, and transition state (TS) of the Diels-Alder reaction

	cyclopentadiene	ethene	norbornene	Transition state
E_e (a.u.)				
$H_\mathrm{t,r,v}$ (kcal/mol)				
$S_\mathrm{t,r,v}$ [cal/(mol·K)]				
G (kcal/mol)				
ΔE_e (kcal/mol)	—	—		
ΔG (kcal/mol)	—	—		

Determination of reaction energy / free energy

First let's determine the reaction free energy: $\Delta G_\mathrm{rxn} = G(\mathrm{P}) - G(\mathrm{R})$. The equation that we employ to calculate the Gibbs free energy of each species is as follows:

$$G = E_\mathrm{e} + H_\mathrm{t,r,v} - TS_\mathrm{t,r,v} \tag{2}$$

where E_e is the electronic internal energy of a molecule, $H_\mathrm{t,r,v}$ and $S_\mathrm{t,r,v}$ are translation, rotation, and vibration's contributions to *enthalpy* and *entropy*, respectively, and T is the temperature (use 298.15 K here). To obtain these quantities, we need to perform harmonic frequency calculations at the *optimized* structures of the reactant species (cyclopentadiene and ethene) and the product (norbornene):

- Build the structure of *cyclopentadiene* using IQmol. Be sure to check the number of hydrogen atoms and use IQmol's energy minimizer ("`E↓`") to pre-optimize the structure.
- Set up geometry optimization followed by vibrational frequency calculation at the B3LYP-D3(BJ)/def2-SVP level. The procedure is the same as how you set up the "OptG" frequency calculations in **Exercise 1**. Again, don't forget to add the "–D3(BJ)" correction.
- You need to check the output text file to get all the numbers you need (or you can use the "`grep`" command): E_e can be found by searching for "`Final energy`" (keep **6 digits** after decimal point), and the values of $H_\mathrm{t,r,v}$ and $S_\mathrm{t,r,v}$ can be found near the very end of the output after "`Total Enthalpy`" and "`Total Entropy`". (**Note:** for this practice, let's use the values *without* "QRRHO" in the front.)
- Repeat the same procedure for *ethene* and *norbornene*. When building the structure of norbornene, make sure that the bridge carbon is placed out of the 6-member ring.
- Save the optimized structures as "`cyclopentadiene.xyz`", "`ethene.xyz`", and "`norbornene.xyz`", which will be used later and also needs to be submitted.

After collecting the data for the reactant and product species **(0.75 pt)**, calculate the total free energy (in kcal/mol) for each species at 298.15 K using Eq. (2), and then calculate the values of $\Delta E_\mathrm{e,rxn}$ and ΔG_rxn for this D-A reaction. Report the numbers in **Table 3**. **(0.5 pt)**

Note: (i) 1 a.u. = 627.5095 kcal/mol; (ii) the unit for entropy reported in Q-Chem output is in cal/(mol·K) rather than kcal/(mol·K).

Determination of transition state and reaction barrier

Up to this point, you should have already completed **columns 2–4** in **Table 3**. Let's now locate the transition state (TS). The D-A reaction involves the formation of two new bonds. As mentioned in lecture, at the TS the length of the newly formed bonds should be elongated by 40–50% compared to equilibrium bond lengths. If we use the lower bound (40%), then the estimated C$\cdots$C distance at the TS should be close to 2.2 Å. Based on this estimation, let's perform a *constrained geometry optimization* first to generate a good guess structure for TS search:

- Use IQmol to open the optimized geometry of norbornene ("`norbornene.xyz`"). Identify the two C−C bonds that are formed in the D-A reaction (whose bond lengths should be close to 1.57 Å) and **delete them** (simply select the bonds and click the red "`X`").

- Set up the bond length constraints: Select the two C atoms originally connected by one of the bonds, then click "`Build` → `Set Geometric Constraint`". On the pop-up window, you should choose "`Constrain`", and then enter the value "2.2 Å". Repeat the same for the other C−C bond. After setting up the constraints, your molecule should look similar to the left panel of **Figure 2**.

  Note: You may notice that the two C$\cdots$C distances are not identical in the end, and one of them is not exactly 2.2 Å. This is because the two constraints are not set up synchronously. There is no need to worry about this since it will not affect the result of constrained optimization.

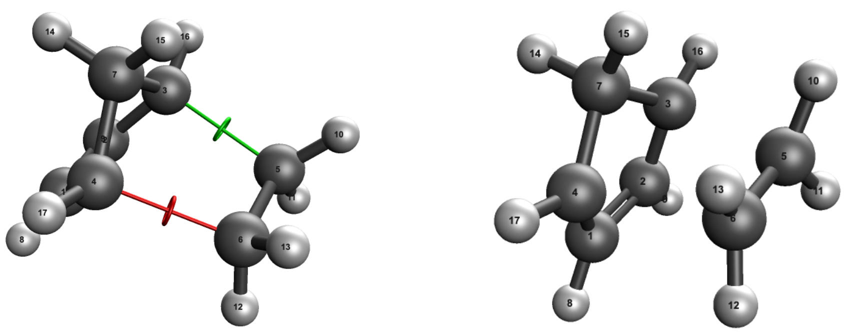

Figure 2: IQmol view of (i) the norbornene structure after geometry constraints are set (left) and (ii) resultant structure of the constrained optimization.

- Go to "`Q-Chem Setup`". Then set "`Calculate` → `Geometry`". Once you did that, the following lines should appear in the generated input file:

```
$opt
CONSTRAINT
stre  3  5    2.2
stre  4  6    2.2
ENDCONSTRAINT
$end
```

  This indicates that the geometry constraints are ready in place. **Note:** (i) The indexes of the bond atoms may be *different* in your molecule, depending on how the norbornene molecule was built initially; (ii) with some versions of IQmol, once you select "`Calculate` → `Geometry`", you may see duplicated "`$opt`" and "`$end`" produced in the generated input. If that happens, please delete the extra ones (otherwise the job will crash).

- Now you just need to set up the correct `Method` and `Basis` to use (B3LYP-D3(BJ)/def2-SVP), then execute the calculation on the server.

After the constrained optimization is finished, you should confirm that the lengths of the two C−C bonds are indeed 2.2 Å. Save the XYZ coordinates obtained as "`DA_constrained.xyz`". You should expect to see a structure similar to the right panel of **Figure 2**. This structure should serve as a good guess for the TS search. We will now set up the TS search job followed by a vibrational frequency analysis:

- Open "`DA_constrained.xyz` and go to "`Q-Chem Setup`". Click "`Reset`" to ensure that the correct structure is loaded and to clean up the input sections.
- Set the job type to "`TS`" (`Calculate` → `Transition State`), and the method and basis set should be set in the same way as above. Note that with Q-Chem 6.1 and after, the "TS" jobs will compute the *exact Hessian* at the beginning by default and read in that as the initial guess Hessian. Therefore, one will *no longer* need to run a frequency calculation at the beginning and read in the Hessian from that.
- To verify that you have successfully located a TS, you will typically need to perform a vibrational frequency analysis. Now add a second job after the TS search and set the job type to "`Frequency`". Make sure that the method and basis are set correctly. Now, execute this job on the Itcotck server. Since the constrained optimization provides a good guess structure, the TS search should converge quickly. The Hessian/frequency calculations are costly though, so the entire job may take a few minutes.
- Once the calculation finishes, check the vibrational frequencies first. Based on the number of imaginary frequencies, is the obtained structure a valid TS? **(0.25 pt)**

  Then double-click on the first imaginary mode to visualize the vibration. **Describe the motion using text below.** Based on the visualization, is this imaginary mode relevant to the D-A reaction? **(0.25 pt)**

  Once the TS is validated, save its structure as "`DA_TS.xyz`". **Submit the five XYZ files** you have saved so far with your report. **(0.25 pt)**
- Report the electronic energy (E_e) of the optimized TS structure and evaluate the reaction energy barrier in **Table 3**: $\Delta E_\mathrm{e}^\ddagger = E_\mathrm{e}(\mathrm{TS}) - E_\mathrm{e}(\mathrm{reactants})$ **(0.25 pt)**
- From the output of the frequency calculation at the optimized TS structure, find the enthalpy and entropy corrections and report them in the last column of **Table 3**. Calculate the Gibbs free energy of the TS and the activation free energy barrier of this reaction at 298.15 K: $\Delta G^\ddagger = G(\mathrm{TS}) - G(\mathrm{reactants})$. Fill your results in **Table 3**. **(0.25 pt)**

Based on your computational results in **Table 3**, sketch a **reaction free energy diagram** below, in which you should use 3 *horizontal lines* to represent reactant (R), transition state (TS), and product (P), and 2 *double-headed arrows* to indicate ΔG_rxn and $\Delta G^\ddagger$. **(0.5 pt)**

Answer the following questions based on your DFT calculations:

1. Determine the value ΔH_rxn of this reaction, which should include contributions from E_e and $H_\mathrm{t,r,v}$. Is this reaction endothermic or exothermic? **(0.3 pt)**

2. Describe the effect of *entropy* on the spontaneity and rate of this reaction. Then rationalize your answer. **(0.5 pt)** (**Hint:** Consider the change in the number of molecules during the course of the reaction.)

3. The theoretical reference for the reaction energy barrier ($\Delta E_{\mathrm{e}}^{\ddagger}$) of this reaction is **18.3 kcal/mol**. It is known that the B3LYP functional tends to underestimate the reaction energy barrier. Do you observe the same trend here? Answer this question by calculating relative % error. **(0.25 pt)**
   Based on our discussion of the remaining challenges of DFT in lecture, which intrinsic issue of the B3LYP functional is most likely to cause this underestimation? **(0.2 pt)**

Refinement of electronic energies

It is very common practice to refine the energetic results using a higher-level electronic method at the structures optimized using a lower-tier but more efficient method. We are now going to recalculate the values of $\Delta E_{\mathrm{e,rxn}}$ and $\Delta E_{\mathrm{e}}^{\ddagger}$ at the ωB97M-V/def2-TZVPPD level of theory *without* reoptimizing the structures. The overall computational method here can be denoted as "ωB97M-V/def2-TZVPPD // B3LYP-D3(BJ)/def2-SVP".

- Open the saved optimized structure of cyclopentadiene ("`cyclopentadiene.xyz`"). Click "`Reset`" to ensure that the optimized structure is correctly loaded.
- Set up a *single-point energy* calculation with "`METHOD = wB97M-V`", "`BASIS = def2-TZVPPD`". Make sure that "-D3(BJ)" has been *turned off* here. Set `SCF_CONVERGENCE = 8` and "`GUI = 0`".
- Execute this calculation on the Linux server. Then repeat it for the optimized structures of ethene and norbornene, as well as the optimized transition state structure.
- After all 4 single-point jobs finish, you can copy the output files back to your local computer and check them using IQmol. You can also directly find the data (converged SCF total energy) you need by executing the following command on terminal:

  ```
  grep "Total energy" [output]
  ```

  If all 4 jobs contain the same keyword in their job names (e.g., "`wb97mv`"), then you can use command `grep "Total energy" *wb97mv*.out` to get the energies all at once.

Report the energies obtained in **Table 4**, and calculate the $\Delta E_{\mathrm{e,rxn}}$ and $\Delta E^{\ddagger}$. **(0.5 pt)** Compared to the theoretical reference value for the reaction energy barrier (18.3 kcal/mol), is the accuracy significantly improved when ωB97M-V/def2-TZVPPD is employed? Answer this question by calculating the relative % error and comparing that with the previous value. **(0.25 pt)**

Table 4: Refinement of the electronic energies for the D-A reaction using ωB97M-V/def2-TZVPPD

	cyclopentadiene	ethene	norbornene	transition state
E_{e} (a.u.)				
ΔE_{e} (kcal/mol)	—	—		